ZJU-Leaper: A Benchmark Dataset for Fabric Defect Detection and a Comparative Study

Chenkai Zhang, Shaozhe Feng, Xulongqi Wang, and Yueming Wang

***Abstract*—Fabric inspection plays an important role in the process of quality control in textile manufacturing. There is a growing demand in the textile industry to leverage computer vision technology for more efficient quality control in the hope that it will replace the traditional labor-intensive inspection by naked eyes. However, there is an underlying viewpoint in most existing fabric datasets that automatic defect detection is a traditional image classification problem, thus more samples help better, which lacks enough consideration about the problem itself and real application environments. After deep communication with users, we find these facts that an assembly line usually has only a few fixed texture fabrics for a long period, users prefer fast deployment and easily upgradable model to a general model and long-time tuning, and users hope the process of collecting samples, annotating, and deployment affects assembly lines as little as possible. This implies that defect detection is different from popular deep learning problems. Multiple-stage models and fast training become more attractive since users are able to train and deploy models by themselves according to the real conditions of samples that can be obtained. Based on this analysis, we propose a new fabric dataset "ZJU-Leaper". It provides a series of task settings in accordance with the progressive strategy dealing with the problem, from only normal samples to many defective samples with precise annotations, to facilitate real-world application. To avoid misleading information and inconsistency issues associated with the prior evaluation metrics, we propose a new evaluation protocol by experimental analysis of task-specific indexes, which can tell a truthful comparison between different inspection methods. We also offer some novel solutions to address the new challenges of our dataset, as part of the baseline experiments. It is our hope that ZJU-Leaper can accelerate the research of automated visual inspection and empower the practitioners with more opportunities for manufacturing automation in the textile industry.**

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***Impact Statement*—Automatic defect inspection is very important in quality control of the fabric industry by helping manufacturers to identify production problems early, hence improving product quality and production efficiency. Meanwhile, it is able to reduce the high labor cost of manual inspection and boost the productivity of the textile industry. To develop effective mathematical inspection algorithms, the fabric dataset serves as an indispensable component to present a practical application environment and enable fair evaluation for algorithms. This paper proposes a new dataset, called "ZJU-Leaper" designed from a viewpoint of multiple-stage models and fast training, containing threefold novelty: 1) the data collection and organization consider the actual requirements and special characteristics of assembly lines in textile factories; 2) it has several designed task settings in order to meet the different levels of requirements in the practical inspection task; 3) it provides a reasonable evaluation protocol for comprehensive comparisons between different inspection algorithms. The preliminary experiments show that some existing algorithms still cannot reach the satisfying performance by this benchmark, which implies more effort should be made to develop new methods for the real use of automatic defect inspection.**

***Index Terms*—Fabric defect detection, automatic inspection, dataset, benchmark.**

I. Introduction

FABRIC defects correspond to the faults on the manufactured textile surfaces and lead to substandard products. Defect inspection is thus very important in quality control of the textile industry by helping manufacturers to identify early production problems, hence improving product quality and production efficiency. However, the current control process relies heavily on human efforts to inspect the fabric by naked eyes, burdening manufacturers with high labor costs, and low efficiency. Moreover, long-term human inspection can cause severe loss of vision. The textile industry, therefore, increasingly demands the leverage of machine-vision technology to replace the traditional method.

In the meantime, the renaissance of deep learning (DL) [1] has made a major impact on the development of artificial intelligence including computer vision technology. A large growing number of state-of-the-art automated visual systems have been built around deep neural networks, thanks to the availability of several open image datasets, such as ImageNet [2], PASCAL VOC [3], and Microsoft COCO [4]. Among the DL applications, automated inspection is considered as an essential part of the industrial quality control system for the future, despite room for improvement in both system performance and reliability. Recent research efforts have been invested in new vision technologies [5]–[11] for inspection tasks in complex

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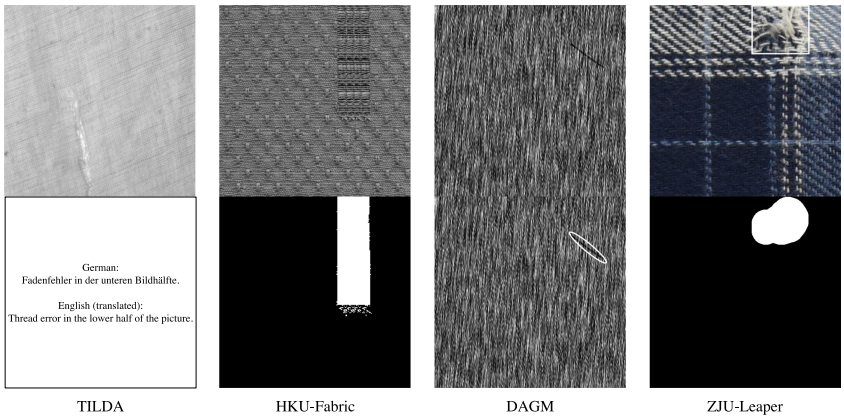


Fig. 1. Images of the defective samples (the first row) and its annotation (the second row) from TILDA, HKU-Fabric, DAGM, and ZJU-Leaper. TILDA roughly describes the location and the shape of defects in the German language. HKU-Fabric is the first dataset that adopts the pixel-wise mask to annotate defective regions. DAGM's images were generated by the algorithm, and the defect region is annotated by an ellipse. The defect of ZJU-Leaper is labeled by a pixel-wise mask and a bounding box. (The additional bounding box annotation of ZJU-Leaper dataset is directly put on the sample image to save space.)

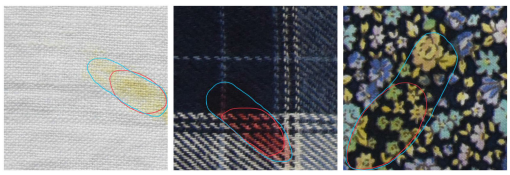


Fig. 2. These three sample images have stain defects. The naked eyes cannot see their boundaries clearly, so different human annotators may define them differently. (Best viewed on screen.)

real-world scenarios using several common textile datasets, such as TILDA [12], DAGM [13], and HKU-Fabric [14]. However, these fabric datasets are insufficient in reflecting the amount, variability, and complexity of real-world fabrics. In addition, the current evaluation methods based on them appear to be inconsistent, making it difficult to enable a fair comparison among them.

The reasons that the existing textile datasets fail to capture the special characteristics and actual requirements of the fabric inspection task can be described as the followings:

- The fabric defects occur occasionally during the massive manufacturing and only occupy a small area of the entire fabric surface. It is therefore difficult to find and collect defective samples in large quantities.
- The fabric defects usually have an unstable and unpredictable appearance in terms of size and structure. In addition, compared to physical objects in real life, the fabric defects often have blurry boundaries and are hard to be defined precisely, which increase the difficulty of obtaining accurate annotations. As shown in Fig. 2.
- In production, fabrics will be changed on demands through the adjustment of raw materials, manufacturing processes, etc. The resulting changes in fabric pattern may cause the failure of detection algorithms, and a new round of image collection and model training is often needed. Thus manufacturers prefer to quickly apply basic inspection models trained using only normal samples or a few defective samples and hope to progressively obtain improved models by providing more defects and precise annotations during the running period of the previous models. But the current datasets do not provide systematic guidance for such problem.
- The fabric defect detection is not the same problem as natural image classification which can be handled by collecting a large amount of data. An effective inspection algorithm should work well on a particular fabric or a group of fabrics with similar patterns, and can be retrained quickly when the pattern changes, instead of a complicated model capable of working for all kinds of fabrics, but needing a long time for tuning and training.

These factors lowered the quality of existing fabric image datasets, and the textile industry demands a new one that can offer a large amount of data with real-world task settings and a unified evaluation protocol. To this end, we introduce a new benchmark fabric dataset, called "ZJU-Leaper". It surpasses the current datasets in terms of sample size, image quality, annotation richness, and more importantly, texture variability and complexity.

The contribution of this paper is three-fold:

Fabric dataset: A new fabric dataset called "ZJU-Leaper" brings two major challenges. The first is the large amount of data. It consists of 98 777 images of fabrics from 19 different fabric categories with detailed annotations. The second is different task settings. It provides a 5-setting origination, from only normal samples to many defective samples with precise annotations, to progressively address the fabric inspection tasks.

Evaluation protocol: A new evaluation protocol with three-level metrics, the pixel-level, the region-level, and the sample-level, are proposed to provide a comprehensive comparison between different inspection algorithms. And all metrics have been proven to work well via meta-measure experiments.

Comparative study: The baseline models in different inspection settings are tested and compared, with the potential of being further extended to future tasks.

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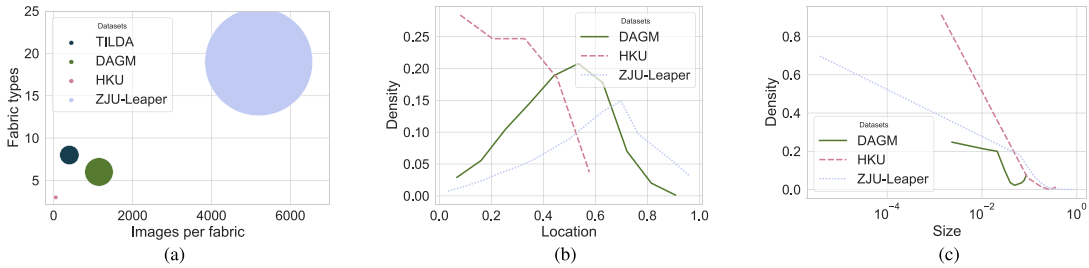


Fig. 3. Analytical comparison between the existing fabric datasets and ZJU-Leaper: (a) the visualization of data volume based on the three aspects: the number of images (circle area), the number of fabric types (y-axis), and the number of images per fabric (x-axis); (b) location distribution of defects: x-axis denotes the distance from defect center to image center, normalized by half the length of image diagonal; (c) size distribution of defects: x-axis denotes the proportion of the defect region to the image size. (TILDA dataset is excluded due to the lack of precise annotations.)

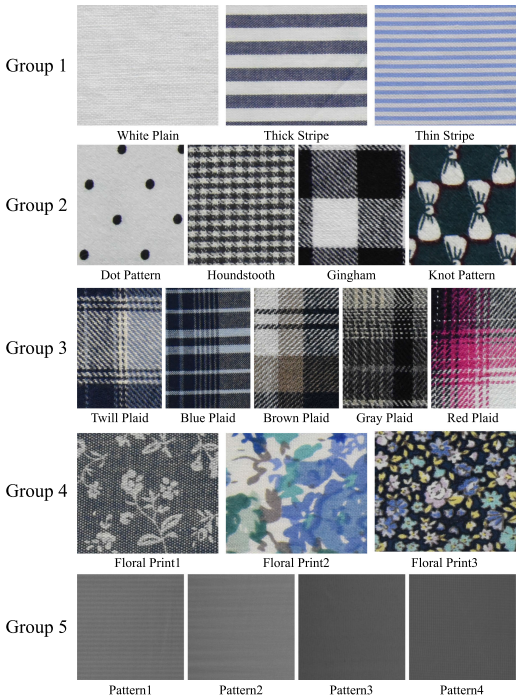


Fig. 4. Example images for 19 patterns and 5 groups. The name of each pattern is indicated in the caption below the pictures. Best viewed on screen.

In aggregation, our works lay a foundation for the future development of a robust open fabric dataset, which will hopefully promote the advancement of automated fabric inspection technology. [1]

II. Related Work

A. Fabric Datasets

There are 3 fabric datasets that have been previously released for fabric defect detection. The first three rows of Table I detail their specifications, respectively.

- TILDA (Textile Texture-Database) [12], released on the Internet in 1996, is the first textile dataset available, focusing on fabric texture research. It contains a total of 3200 images from 8 different texture groups. Each group contains 1 normal class and 7 defect classes wherein text annotations only roughly describe the location and the shape of defects in the German language (1st column of Fig. 1). The quality of the images, mostly taken over 20 years ago, is low. In addition, a fee is required to access TILDA.
- DAGM (Deutsche Arbeitsgemeinschaft für Mustererkennung) 2007 [13] is a public dataset created in 2007 for an industrial image processing competition — the Weakly Supervised Learning for Industrial Optical Inspection. It contains approximately twice the number of images in TILDA but fewer images of defects. In fact, each texture group has 1000 images of normal textile and only 150 pictures of defects. And all images of DAGM were artificially generated by the computer program, which makes them less resemblant to the real-world samples, hampering their value for application. The defect region is annotated by an ellipse with the center point coordinate and the lengths of the major and minor axes (3 rd column of Fig. 1), which in general is more accurate than rectangle annotation.
- HKU-Fabric [14] was constructed by the Industrial Automation Research Laboratory of the University of Hong Kong in 2009. It divides fabrics into three types: box-, star-, and dot-patterned, containing 56, 56, and 50 images, respectively. Some of the images are generated from the simulation. This dataset adopted the pixel-level mask for the first time to annotate defective regions (2nd column of Fig. 1), which can provide more precise information for defect definition and performance evaluation. HKU-Fabric has therefore become a popular dataset for fabric inspection tasks.

B. Inspection Methods

Most traditional fabric inspection algorithms can be mainly classified into 4 categories, namely, statistical, structural, spectral, and model-based algorithms [15], [16]. Statistical approaches measure the spatial distribution of pixel values, using histogram statistics [17], co-occurrence matrices [18], mathematical morphologic [19], or other tools [14], [20]. Structural approaches, on the other hand, regard fabric pattern as a composition of texture primitives [21] and placement rules [22],

[1]The dataset will be public when this paper is accepted. And the related materials are available at: http://www.qaas.zju.edu.cn/zju-leaper and https://github.com/nico-zck/zju-leaper-dataset

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TABLE I
COMPARISON BETWEEN EXISTING PUBLIC FABRIC DATASETS AND ZJU-LEAPER

	Images					Annotations			Others			
	Fabrics	Defective	Normal	Total	Color Space	Label	BBox	Mask	Pattern groups	Test set splitting	Evaluation protocol	Factory data
TILDA [12] (1994)	8	2798	400	3198	grayscale	✓						
DAGM [13] (2007)	6	900	6000	6900	grayscale		✓					
HKU-Fabric [14] (2009)	3	81	85	166	grayscale			✓				
ZJU-Leaper (ours)	19	27650	71127	98777	RGB	✓	✓	✓	✓	✓	✓	✓

[23]. Meanwhile, spectral analysis algorithms use Fourier transform [24], [25], wavelet transform [26], [27], Gabor transform [28], [29], or Contourlet transform [30] to discover texture periodicity in the frequency domain. Lastly, model-based algorithms represent texture characteristics as the parameters of certain statistical models including Gaussian mixture models (GMM) [31] and Markov random field (MRF) [32]. These approaches were designed for plain and twill textures and usually obtain poor performance on complex fabric images. In fact, automated fabric inspection remains challenging and falls short in practical applications, particularly when taking human performance as a reference.

In contrast to the conventional methods, some methods such as Deep Learning (DL) and Support Vector Machine (SVM) are utilized to inspect fabrics of complex patterns. These methods can be roughly divided into two groups: one-class learning and multi-class learning. The multi-class (including binary-class) method requires both defect-free and defective samples for training. A CNN-based architectures, such as VGG [33] and U-Net [34], can be directly applied to perform fabric inspection [5] and classification [35]. The disadvantage, however, is that a large number of defect samples of certain fabric must be collected for training purposes, which is especially difficult because the defects occur rarely.

On the other hand, one-class approaches, such as some anomaly detection methods, only require normal data for training. For example, a one-class SVM classifier [36] can distinguish outliers based on decision boundaries learned from normal samples. And Zhang *et al.* [9] used CNN to map normal samples into a certain *feature space* prior to performing one-class anomaly detection. Another approach to the one-class method is through reconstruction-based processes. Basic auto-encoders [6]–[8] are used to regenerate an input after its passing through certain bottleneck layers. *Dictionary learning* methods [37], [38] have also been shown to reconstruct images from the restricted representation of normal samples encoded in its dictionary. However, these methods tend to compress image content without fully grasping its semantically meaningful information. Recently, Schlegl *et al.* [11] used Generative Adversarial Networks (GANs) [39], known for producing photorealistic images, to improve the quality of regenerated fabric images. But it is relatively slow due to the time-consuming optimization process.

In addition, some unconventional but interesting methods have been proposed for defect detection in recent years, inspired by technologies used in image decomposition [40], game theory application [41], and lattice structure [42], [43]. The low-rank decomposition with graph-based segmentation method [44] achieved state-of-the-art result on the HKU-Fabric. At the same time, some studies adopt thermal images to detect both surface and inner defects [45]–[48]. However, these methods are designed for simple patterned fabrics, such as those in the TILDA, HKU-Fabric, and DAGM, and will have limited performance on fabrics with complex textures.

III. ZJU-LEAPER DATASET

Designing a benchmark dataset requires much consideration in data preparation, image collection, annotation methodology, and evaluation protocol. In this section, we describe the ZJU-Leaper dataset in four aspects: the collecting and labeling process of dataset construction, the analysis and comparison of dataset attributes, the dataset organization for different task settings, and the new protocol to evaluate the performance of inspection algorithms.

At the same time, two major challenges emerged. The first is related to the amount of data, which requires the inspection algorithms to have the ability to process large amounts of data with varying fabric patterns, but most of the existing methods fall short of that expectation. The second is related to task settings. Different task settings will result in different costs for data collecting and labeling, in which a progressive solution could be used to address the automated inspection tasks. It is challenging to build an effective detection model by only normal samples or a small number of defective samples. And training an inspection model without precise defect region information is also a difficult task. Basically, the less defect data involved, the more challenging the task.

A. Dataset Construction

1) Image Collecting: The first stage of the dataset construction involves collecting candidate images for each fabric type. The number of images per pattern in existing datasets [12]–[14] is usually no more than 1000, whereas ZJU-Leaper aims to offer much more. We first built up a huge set of candidate images, either collected from the factory or taken in the laboratory. After screening and adjustment, each fabric type contains around 5000 images. Those from the laboratory were captured in a stable environment with an identical camera setting and a fixed illumination condition. The high resolution and full-color format were adopted to save more texture details.

At this point, the vast majority of images in the dataset were of normal fabrics. We then randomly selected normal fabric images three times the number of defect samples for each pattern to reduce data redundancy. As a result, ZJU-Leaper acquired a

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TABLE II
STATISTICS OF EACH FABRIC IN THE ZJU-LEAPER DATASET

	Group 1			Group 2				Group 3					Group 4			Group 5 (Factory)				
	White Plain	Thick Stripe	Thin Stripe	Dot Pattern	Houndstooth	Gingham	Knot Pattern	Twill Plaid	Blue Plaid	Brown Plaid	Gray Plaid	Red Plaid	Floral Print1	Floral Print2	Floral Print3	Pattern1	Pattern2	Pattern3	Pattern4	**Total**
Defect	891	1026	1026	1026	1362	1389	1525	1423	1493	1467	1169	1326	1312	1232	1177	1367	3177	2863	1399	**27650**
Normal	2673	3080	3080	3080	4086	4167	4575	4269	4479	4401	3507	3978	3936	3696	3531	3190	5433	1769	4197	**71127**
SUM	3564	4106	4106	4106	5448	5556	6100	5692	5972	5868	4676	5304	5248	4928	4708	5248	4928	4708	4708	**98777**

total of 98 777 high-quality images of 19 fabric patterns. The statistical information of the ZJU-Leaper dataset can be found in Table II and some visual examples can be found in Fig. 4.

2) Dataset Annotation: We rely on humans for accurate annotation of the candidate images collected in the previous step. This laborious task was achieved by 6 persons with normal visual functions. During the process, the annotators were asked to first identify whether there was any defect in an image, and secondly label the silhouette of it if existed.

However, there are two issues to be considered. Firstly, the annotation requirements may vary according to different inspection tasks. In order to meet versatile needs, we offer three different types of annotations: image label, bounding box, and pixel-wise mask. All three forms are commonly used in the computer vision community and easy to obtain via the existing annotation tools. In the annotation information of ZJU-Leaper, each fabric image has its own labels to show whether it is defective, its pattern name, pattern number, group name, and group number (the group is introduced in Sec. III-C1). Secondly, as Fig. 2 shows, the generally unclear defect outline increases the difficulty of accurate annotation. Our solution to this issue was to have three annotators label the same image independently (except for the annotated images acquired from the factory). We then assembled the final mask by averaging their outputs.

B. Dataset Analysis

1) Dataset Volume: Obtaining a considerably large number of images is essential to capture the diversity and richness of real-world fabrics. Our dataset consists of 27 650 defect images and 71 127 normal ones. Fig. 3(a) intuitively shows the comparison of data volume between ZJU-Leaper and other fabric datasets. As it indicates, ZJU-Leaper accounts for the largest dataset with respect to the number of fabric samples, fabric types, as well as images per type.

2) Defect Location: Center bias has been identified as one of the most significant biases occurring during detection tasks [49]. To analyze this phenomenon, we plotted the distribution of defect location for various datasets in Fig. 3(b), where the x-axis denotes the distance between a defect center and its image center, normalized by half the length of the image diagonal. Compared with other fabric datasets, ZJU-Leaper exhibits the least center bias. Of note, we excluded TILDA [12] dataset from this exercise due to its lack of workable annotation.

3) Defect Size: The size of the defect is represented by the proportion of defect pixels in an image. Fig. 3(c) shows that the defect size of ZJU-Leaper varies across a much broader range, compared with HKU-Fabric [14] and DAGM [13]. Again, TILDA [12] was not analyzed.

C. Dataset Organization

1) Pattern Groups: Unlike most algorithms applied to natural images, fabric defect detection methods are usually texture-sensitive and an inspection algorithm is usually applicable to a single fabric or several fabrics with closely related fabric patterns. To help users easily access the relevant data and algorithms, we organized the ZJU-Leaper dataset into five major texture groups with each group containing 3-5 similar patterns, as listed in Table II and shown in Fig. 4. Specifically, we grouped fabrics of simple patterns (e.g., plain and striped ones) into Group 1. Group 2 consists of those with small repeated motifs (or primitives), such as checks and dots, which show a clear visual arrangement. Group 3 and Group 4 are composed of plaid and floral fabrics, which have complex arrangements or complex patterns. And finally, four greige fabrics (neither bleached nor dyed) were collected from the factory making up Group 5. The data in this group are of poorer quality in terms of image resolution, illumination consistency, and annotation accuracy. They are included in our study solely for comparison purposes. In addition, tags of group name and group number are added to the annotations of each image.

2) Task Settings: In production, the texture of fabric under inspection can vary and might not fall into any of our five pattern groups. It is necessary to first collect a sample amount of training data, based on which an inspection model can be built. However, collecting defect data can be both costly and time-consuming for the reasons mentioned earlier. In addition, different types of annotation require varying amounts of effort. For instance, the workload of mask labeling is about 15 times heavier than that of object localization according to the report of [4]. We therefore propose a 5-setting organization of training data to solve the fabric inspection tasks via a progressive manner. With every setting further, it requires more costs and human efforts but rewards with more defect data for training. Basically, the less defect data the model uses, the more difficult the task setting becomes. Based on these settings, the inspection tasks can be solved from coarse-grained annotations or even unsupervised learning, and then finer-grained annotations are used to improve the detection performance until a satisfactory performance level is achieved. The five settings can be described as below:

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- *Setting 1:* Normal samples only (annotation-free) — when images of defect-free fabrics can be obtained cheaply, and those of defects are difficult to acquire.
- *Setting 2:* Small amount of defect data (with mask annotation) — when a small amount of defect data and annotation information can be obtained at a low cost, along with many normal images.
- *Setting 3:* A large amount of defect data (with label annotation) — when methods for the first two settings fail to train the inspection model effectively, and labeling information of defects can be quickly obtained.
- *Setting 4:* A large amount of defect data (with bounding-box annotation) — more labor is involved in marking defects with bounding boxes that contain rough position and size information of defects.
- *Setting 5:* A large amount of defect data (with mask annotation) — used as the last resort where the effort to mask all defects in a large number of samples becomes necessary.

In summary, Setting 1 and Setting 2 are proposed since fabric defects do not appear as usual as normal fabrics in assembly lines. Annotations are expensive to obtain since the workers cannot always stop working for a long time to help label defects for short-lived fabrics, thus there are Setting 3 and Setting 4. Finally, Setting 5 with detailed annotations are provided, in order to enable researchers to have enough information to analyze the algorithms' behavior on Settings 1 to 4, which is important for further improvements. Obviously, the most ideal but challenging way to obtain the inspection model is to train it on Setting 1, which only requires some normal samples. On the contrary, obtaining the model on Setting 5 is most expensive because it requires a large amount of defect data.

3) Test Data: Unfortunately, none of the existing public fabric datasets clearly define the test data used for performance evaluation of inspection methods. As a result, these datasets were used by previous works variously without a uniform test data baseline. It is hence hard to identify the true high-quality algorithms. To avoid the same problem, we pre-defined one test set for each pattern group in ZJU-Leaper to ensure uniform evaluation and fair comparison between detection methods. These test data sets were obtained by randomly selecting 1/3 of the data, including both normal and defect samples, from each pattern group. After obtaining training data in one of the above-mentioned five settings, users are required to use the corresponding test data set for model evaluation. It must be emphasized that the usage of any data in the test set during model training and hyper-parameter selection is strictly prohibited.

D. Evaluation Protocol

Before introducing the metrics of our evaluation protocol, there are some important things to note. First, the evaluation metrics used in related work are usually inconsistent, which makes existing methods difficult to compare with each other. Second, the scanty number of defective pixels causes a data imbalance problem. Hence the pixel-wise accuracy, specificity, and false alarm widely used in previous work [29], [40], [41], [50], [51] will bring misleading information. Third, pixel-level metrics will mix up all pixels together while the information of each single defect region is overlooked. To solve these problems, we will present a new evaluation protocol for fabric inspection methods. In addition, some of the existing methods involve the classification of defect types which, unfortunately, follows all kinds of standards without an industrial consensus. To avoid importing such confusion, we focus our evaluation protocol solely on the detection function of inspection algorithms, rather than the defect classification capability.

Concretely, our evaluation protocol measures the performance of detection algorithms from three different levels, the pixel level, the region level, and the sample level. In the end, some meta-measure experiments suggest that the metrics used in this protocol are capable of evaluating inspection models. And the codes of the evaluation protocol program are publicly available online.

1) Pixel Level: To setup notations, we consider defective sample as positive and normal sample as negative.Let TP be the number of true positives, FP be the number of false positives, TN be the number of true negatives, and FN be the number of false negatives. TP^{pix} and TN^{pix} represent the number of pixels that the prediction matches the annotation, FP^{pix} and FN^{pix} represent the number of pixels that the prediction mismatches the annotation, respectively.

There are many metrics for evaluating pixel-wise results. Among them, Intersection over Union (IoU) and Dice coefficient (Dice) are the most two popular metrics. The Dice coefficient (also known as pixel-wise F1-score) was proposed to evaluate the quality of medical image segmentation by Zijdenbos *et al.* [52]. And the Intersection over Union (IoU) is widely used in semantic segmentation for natural image datasets (e.g. VOC [3] and COCO [4]).

$$IoU = \frac{TP^{pix}}{TP^{pix} + FP^{pix} + FN^{pix}} \tag{1}$$

$$\begin{aligned} \boldsymbol{Dice = F_1^{pix}} &= \frac{\boldsymbol{TP^{pix}}}{\boldsymbol{TP^{pix} + \frac{1}{2}(FP^{pix} + FN^{pix})}} \\ &= \frac{\boldsymbol{2 \cdot IoU}}{\boldsymbol{IoU + 1}} \end{aligned} \tag{2}$$

However, from 2, we found IoU could be derived from the Dice, which means that the IoU score is just a similar representation of the Dice coefficient from the mathematical view. Hence using only one of them is sufficient for pixel-level evaluation. We prefer Dice coefficient since IoU is usually used to evaluate the detection quality of a single object region [3], [4], while Dice is usually used for an overall pixel-wise evaluation [52], [53]. More importantly, it achieves better alignment with perceptual quality under the class-imbalanced scenario [52]. Some metrics, such as accuracy, specificity, and false alarm are excluded from this protocol as they do not apply well to imbalanced data.

2) Region Level: In the pixel-level evaluation, all pixels are mixed for calculating the metrics. Therefore, at least three sources of information are overlooked. The First is spatial information. As Fig. 5 shows, pixel-level metrics cannot tell the differences between these defect regions at different locations.

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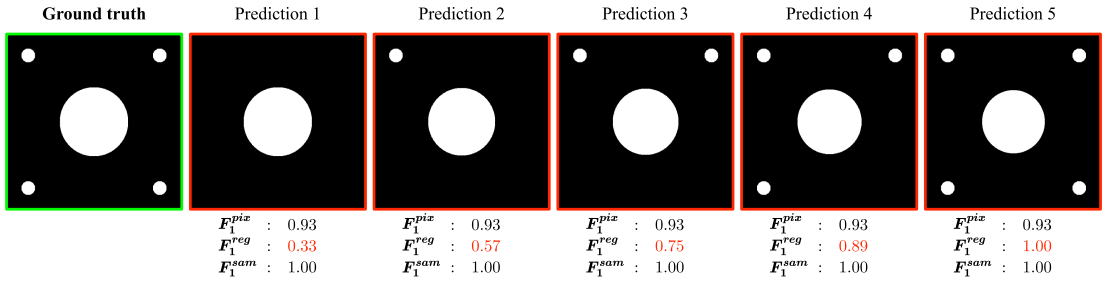


Fig. 5. A toy example to illustrate that when considering the spatial information of defects in the ground truth and predicted masks, pixel-level metrics (the first line of the results below images) cannot capture the differences between these defect regions, which suggests the use of region-level metrics (the second 2 lines) for evaluating is necessary. The target image in the first column (green border) is composed of 1 large circle and 4 small circles. These 5 predicted masks (red border) change as the number of small circular defects increases. Note that all these predictions have the same number of "white" pixels.

The second is the shape information of defects. Inspection models are likely to predict discrete masks for the defects with unusual shape (e.g., too large or too slender) or fuzzy boundary, while human annotators tend to label the adjacent defects as a whole part, all of which make it difficult to establish a one-to-one correspondence between the ground truths and the predictions. The third is size information. Evaluating inspection methods on defects of different sizes could provide us with insightful analysis. However, the variety of defects in ZJU-Leaper makes the defects with different sizes sometimes appear in the same picture, and pixel-level evaluation cannot recognize the size of each defect.

To resolve these problems above, we first generate proposed regions and ground-truth regions through isolated pixel clusters in mask predictions and mask annotations respectively and then establish a one-to-one or many-to-one relationship between proposed regions and ground-truth regions. Then we introduce IoU (1) and the area ratio (ARR) as criteria to determine whether a proposed region is correct or a ground-truth region is detected. The former quantifies the percent of the overlap and the union between a ground-truth region and a proposed region, and the latter represents the ratio of the overlap to their own area. Such criteria relax the requirement of only using IoU that a proposed region must be close to the ground-truth region even in the many-to-one condition.

Finally, the detection performance then is evaluated by region-level precision and recall, in which the F1-score is used to aggregate them robustly.

$$Pre^{reg} = \frac{\#correct\ proposed\ regions}{\#total\ proposed\ regions} \tag{3}$$

$$Rec^{reg} = \frac{\#detected\ ground--truth\ regions}{\#total\ ground--truth\ regions} \tag{4}$$

$$\boldsymbol{F_1^{reg} = 2 \cdot \frac{Pre^{reg} \cdot Rec^{reg}}{Pre^{reg} + Rec^{reg}}} \tag{5}$$

where counting a correct proposed region or a detected ground-truth region is based on the following requirements: IoU is greater than 0.3 or ARR is greater than 0.51 (more than half of a proposed/ground-truth region). Of note, since defects in fabric images are usually scattered and surrounded by normal texture, certain position offset does not significantly impact the detection results. Therefore, a lower threshold of IoU should be adopted to increase the fault tolerance to biased annotations.

In addition, based on the region-level metrics, the evaluation for different sized defects could be readily obtained when adding size constraints to region-level precision and recall:

$$F_1^{reg}{}_{\{s,m,l\}} = 2 \cdot \frac{Pre^{reg}_{\{s,m,l\}} \cdot Rec^{reg}_{\{s,m,l\}}}{Pre^{reg}_{\{s,m,l\}} + Rec^{reg}_{\{s,m,l\}}} \tag{6}$$

in which subscripts $\{s, m, l\}$ represent small, medium, and large size constraints. Then the calculating of $Pre^{\{s,m,l\}}$ and $Rec^{\{s,m,l\}}$ will only account for the ground-truth regions that satisfy the size constraint. The details about choosing appropriate size constraints for ZJU-Leaper are discussed in Appendix A.

3) Sample Level: Large defect regions will dominate the value of pixel-level metrics. Similarly, only region-level evaluation leads to a bias situation if there are many defect regions on one image being successfully found and the other images only have a small number of such regions missed. Thus, we also propose a sample-level metric to evaluate the performance over different fabric images.

$$F_1^{sam} = \frac{TP^{sam}}{TP^{sam} + \frac{1}{2}(FP^{sam} + FN^{sam})} \tag{7}$$

where TP^{sam} represents the number of matched pairs of sample prediction and the label, FP^{sam} and FN^{sam} represent the number of unmatched pairs.

4) Meta-Measure Experiments: Proving the quality of evaluation metrics is a difficult task [54]. Inspired by the work of Pont-Tuset *et al.* [54], [55], we adopted the meta-measure (a measure that evaluates evaluation metrics) to verify the effectiveness of these metrics. Concretely, we employ three meta-measure experiments, based on the following properties:

- The metric score of a mask prediction should be low when using incorrect ground truths. And as the ratio of using incorrect ground truths increases, the score should decrease. (Fig. 6(a)).
- A metric should have the ability to tell the difference between using original mask and using shuffled mask as a ground truth (Fig. 6(b)).

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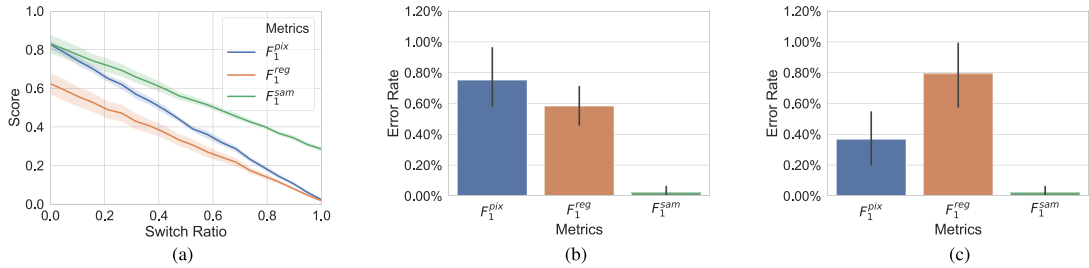


Fig. 6. The results of meta-measure experiments for three levels of metrics in the evaluation protocol. A U-Net based model trained on Setting 5 is selected as the best baseline method. (a) This figure represents how these scores change (y-axis) when evaluating the best baseline method using a certain ratio of switched ground-truths (x-axis). It can be clearly seen that when more and more shuffled ground truths are used, all metrics have a stable downward trend. And a 95% confidence intervals are also drawn around the lines; (b) Meta-measure of switching ground truths. It records the percent of times that a metric wrongly scored a higher value when using a shuffled mask as the ground truth. The lower the score, the better. Our metric achieved reasonable performance with a low error rate. (c) Meta-measure of the best method versus a generic algorithm (outputs centered Gaussian circles independent of the image content). It records the percent of times that a metric graded a higher score to the generic output rather than the result of the best baseline method. The lower the better. Our measure achieved reasonable performance with low error rates. The bar of the sample-level metric is very low in (b) and (c) as pixel-wise alterations are not likely to affect sample-level outputs.

- A metric should prefer the results of good models rather than an algorithm that generates arbitrary masks without considering the content of the image (Fig. 6(c)).

As Fig. 6 (a) shows, there is a reasonable relationship between the value of these metrics and the ratio of incorrect ground-truths. The error rates of F_1^{pix}, F_1^{reg}, and F_1^{sam} are all lower than 1% (in 6 (b) and (c)), which are fairly good according to the reports of [54], [55]. The bar of F_1^{sam} is very low because pixel-wise alterations are unlikely to affect sample-level outputs. In summary, the results of meta-measure experiments indicate that the proposed evaluation metrics are reliable proxies for the performance of inspection models.

IV. Baseline Experiments

In this section, we conducted experiments by applying various detection algorithms on the ZJU-Leaper dataset in 5 task settings, some of which are first introduced to solve fabric inspection tasks. And the performance results are evaluated by the proposed evaluation protocol. Therefore, the main goals of these baseline experiments are to demonstrate the potential inspection solutions, rather than nominating the best method. And we believe these tests could inspire future research on the ZJU-Leaper dataset. All codes and other details could be found on the dataset website.

As mentioned in the introduction, an effective inspection should work on a particular fabric or a group of fabrics with similar patterns. So the following experiments obtain models by training on each fabric and organized the results by the pattern groups.

A. Setting 1: Normal Samples Only

As mentioned earlier, normal fabric samples with stable texture attributes are much easier to collect than defective ones. It is therefore more feasible to start with Setting 1 for defect inspection on new fabric, that is, to collect a large number of normal samples for model training. In fact, some methods of this kind have been proposed [7], [8], [37], [38] to capture the regularity of texture attributes from normal data and then distinguish defects by their texture inconsistency. The advantage of this method, besides the easy access to normal data, is that it can detect defects unassociated with any texture in the training set. However, most of the existing methods relevant to this scenario are designed for plain homogeneous textures, and the data-driven supervised methods requiring a large number of well-labeled images are not applicable here.

In this experiment, we tested three workable solutions. Their performances are then measured by the proposed evaluation protocol, whereof the results are detailed in Table III. The implementations of other typical methods applicable to this setting [14], [40], [41], [43], [44] are listed in Appendix A.

1) Sparse Coding (SC): As a method of *dictionary learning*, sparse coding aims at representing data as a sparse combination of dictionary atoms, which is reportedly similar to how the mammalian primary visual cortex works. Sparse coding was applied in [37], [38] on image patches for the machine to learn a compact dictionary, from which the normal images can be reconstructed well while the defective ones will give outstanding reconstruction errors instead. In this experiment, we implemented a *sparse coding* [37] algorithm to reconstruct fabric images, which uses an orthogonal matching pursuit (OMP) algorithm to address the sparse approximation problem. Specifically, we first use histogram equalization to preprocess grayed images and then extract patches from training and testing images with a size of 128 and a stride of 64. The number of dictionary elements is set to 128 for all fabrics during the experiment.

2) Convolutional Auto-Encoder (CAE): The approach of CAE [7], [8] in general is more capable of common feature extraction and reconstruction for normal samples. Instead of using the learned dictionary in sparse coding, CAEs try to limit the representation of normal textures in a low-dimensional hidden space by passing data through bottleneck layers. In this experiment, we adopted the CAE model [8] with a 4-layer convolutional encoder and decoder. An Adam solver [56] based mini-batch stochastic gradient descend (SGD) with a batch size of 8 and a learning rate of $1e^{-4}$ was used to optimize the model parameters for a total of 60 epochs.

3) One-Class SVM (OCSVM): Since new and unexpected errors occur constantly, it is impossible to enumerate all defects and obtain a complete representation of all fabric defects [57].

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TABLE III
THE RESULTS OF BASELINE EXPERIMENTS UNDER SETTING 1-5. ALL MODELS ARE EVALUATED ON THE TEST SET OF EACH FABRIC

Setting 1	Group	F_1^{pix}	F_1^{reg}	F_1^{sam}
SC	Group1	0.319	0.212	0.610
	Group2	0.175	0.099	0.434
	Group3	0.226	0.131	0.470
	Group4	0.062	0.030	0.393
	Group5	0.064	0.062	0.432
	Overall	0.169	0.107	0.468
CAE	Group1	0.455	0.221	0.461
	Group2	0.216	0.091	0.410
	Group3	0.095	0.046	0.406
	Group4	0.070	0.041	0.393
	Group5	0.098	0.115	0.490
	Overall	0.187	0.103	0.432
OCSVM	Group1	0.259	0.164	0.595
	Group2	0.110	0.062	0.473
	Group3	0.061	0.042	0.423
	Group4	0.148	0.076	0.466
	Group5	0.013	0.012	0.387
	Overall	0.118	0.071	0.469
Setting 2	Group	F_1^{pix}	F_1^{reg}	F_1^{sam}
U-Net	Group1	0.704	0.409	0.890
	Group2	0.740	0.475	0.880
	Group3	0.544	0.220	0.699
	Group4	0.699	0.362	0.728
	Group5	0.103	0.027	0.358
	Overall	0.558	0.299	0.711
U-Net (TL)	Group1	0.783	0.523	0.886
	Group2	0.720	0.516	0.814
	Group3	0.523	0.377	0.661
	Group4	0.750	0.545	0.812
	Group5	0.238	0.159	0.520
	Overall	0.603	0.424	0.739
U-Net (DA)	Group1	0.674	0.416	0.934
	Group2	0.670	0.437	0.882
	Group3	0.543	0.242	0.653
	Group4	0.719	0.404	0.848
	Group5	0.335	0.176	0.579
	Overall	0.588	0.335	0.779

Setting 3	Group	F_1^{pix}	F_1^{reg}	F_1^{sam}
U-Net (Label)	Group1	0.225	0.119	0.603
	Group2	0.248	0.156	0.693
	Group3	0.302	0.170	0.754
	Group4	0.256	0.064	0.420
	Group5	0.089	0.058	0.465
	Overall	0.224	0.113	0.587
Setting 4	Group	F_1^{pix}	F_1^{reg}	F_1^{sam}
U-Net (BBox)	Group1	0.722	0.652	0.952
	Group2	0.727	0.625	0.864
	Group3	0.676	0.457	0.830
	Group4	0.712	0.440	0.755
	Group5	0.577	0.516	0.801
	Overall	0.683	0.538	0.840
Setting 5	Group	F_1^{pix}	F_1^{reg}	F_1^{sam}
U-Net	Group1	0.944	0.854	0.928
	Group2	0.919	0.619	0.805
	Group3	0.938	0.829	0.929
	Group4	0.961	0.805	0.879
	Group5	0.768	0.516	0.740
	Overall	0.906	0.724	0.856
U-Net (Group)	Group1	0.855	0.839	0.957
	Group2	0.794	0.759	0.896
	Group3	0.770	0.686	0.864
	Group4	0.817	0.781	0.936
	Group5	0.603	0.441	0.687
	Overall	0.768	0.701	0.868
U-Net (Total)	Group1	0.816	0.790	0.936
	Group2	0.773	0.689	0.849
	Group3	0.787	0.724	0.893
	Group4	0.796	0.640	0.812
	Group5	0.561	0.418	0.690
	Overall	0.747	0.653	0.836

As a method of *outlier detection*, OCSVM describes the normal data only and regards all the rest as outliers. In this experiment, the previously trained 4-layer encoder of CAE was used to extract features from image patches with a size of 256 and a stride of 64. And SVM model with radial basis function (RBF) kernel was applied to detect anomalies in the feature space.

As indicated in the top-left of Table III, all three methods exhibited the best performance on Group 1 among all groups, confirming that these unsupervised methods are more effective with simple patterned fabrics. The unsatisfactory performance of these unsupervised algorithms on more complex patterns (Group 2-4) highlights the room for improvement in defect inspection methodology under this low-cost task setting. However, for Group 5, also dominated by simple textures, did not furnish good outcomes. This can be attributed to the noises introduced into this group of data during the image collection process by the factory, including fabric creases, image distortion, illumination inconsistency, and image scale variation. In aggregate, these noises exerted a negative impact on the traditional methods SC and OCSVM, while the CNN-based CAE was relatively less affected.

B. Setting 2: Small Amount of Defect Data

In Setting 2, the cost of collecting a small amount of defect data with pixel-wise annotations is relatively low. The main problem with defect detection in this setting can be regarded as the small data problem in machine learning, and some possible solutions can be adopted to address it. In our practice, *transfer learning* and *data augmentation* are used as potential solutions. Considering the task similarity between cell images and fabric pictures, both featuring homogeneous background and valuing low-level semantic information, we used U-Net [34] as the benchmark model for this and the remaining experiments (Sec. IV-D).

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1) Transfer Learning (TL): Transfer learning could be used to leverage the knowledge learned from old fabrics or cheap unlabeled data to solve the small defect data problem of new fabrics. Inspired by Yosinski [58] and Kim [10], we conducted similar transfer learning strategy on fabric defect detection. But unlike [10] where VGG classifier was extracted from ImageNet dataset, we transferred a 4-skip-connection U-Net model [34] from a pre-designed unsupervised task to the supervised segmentation task with only a limited number of defective samples. Specifically, the pre-designed task requires a model to perform image reconstruction over normal samples, and the encoder of the model was exported for training on the new task. The training epochs of the first and second tasks were 20 and 120. The learning rate and weight decay were set to $1e^{-4}$ and $1e^{-5}$, respectively.

2) Data Augmentation (DA): Many recent studies demonstrated the remarkable performance of data-driven deep learning in various computer vision tasks. However, the scarcity of defect samples prevents the application of this approach in the fabric inspection task. Data augmentation, better known as a solution to reduce overfitting on neural network models, can also effectively alleviate the small data problem through synthesizing artificial defect data [59], [60].

As the assumes of Setting 2 that defects are hard to access, it can only provide a small number of images with defects. In this experiment, we implemented an improved way to both transfer and synthesize defects from a small amount of real defect data and then inject them into the normal images. Concretely, we first used ground-truth masks to extract defect regions by filtering out the background on defect images of the training set. Then random affine transformations and lightness/color perturbations were used to transfer defect regions to defect-free images. Therefore, it allows us to train a fully-supervised model with more defect data to perform the defect detection task under this setting. Specifically, a U-Net with 4 skip connections were used in this experiment. The learning rate, weight decay, batch size, and training epochs were set to $1e^{-4}$, $5e^{-5}$, 16, and 40, respectively.

The bottom-left results of Table III show that using even a small amount of supervision for training can improve the detection performance with acceptable outcomes. Notably, the overall performance of U-Nets with data augmentation (DA) and transfer learning (TL) are better than the plain U-Net trained on Setting 2, which confirms that DA and TL are one of the feasible solutions to address the problem of small defect data. The high scores are no longer limited to Group 1, which could indicate that CNN-based models can extract good features for some fabric patterns with high visual complexity. For factory data (Group 5), the small number of defect samples does prevent the model from obtaining sufficient representation of defects. Fortunately, both transfer learning and data augmentation can significantly alleviate this problem.

C. Setting 3-4: Large Amount of Defect Data With Different Annotations

In Setting 3 and 4, where there are many defect images, the problems of annotation become more outstanding, such as the poor defect definition due to the unclear defect boundaries and the labor-intensive nature of precise labeling. Herein, we try to resort to weakly-supervised image segmentation to solve this problem on a large number of defects with either label annotations (Setting 3) or bounding-box annotations (Setting 4). Their performance results are shown in the middle right of Table III.

1) Label Annotations: The first weakly-learning image segmentation algorithm uses label annotation as supervision [61]. The model considers solely a single label for a given image and weighs more on the important pixels for classification, which can be seen as an example of *Multiple Instance Learning*. During the implementation, Log-Sum-Exp (LSE) and Image-Level Prior (ILP) [61] were used to utilize label annotations for fabric inspection. For every training image, LSE was used to extract image-level labels from the mask outputs of the model, and ILP was used for smoothing the final results. Specifically, a U-Net with 4 skip connections were trained with label supervision. During the training process, the learning rate, weight decay, batch size, and training epochs were set to $1e^{-4}$, $5e^{-5}$, 24, and 60, respectively.

2) Bounding-Box Annotations: The second approach uses the bounding boxes (BBox) as an alternative source of supervision to train the CNN-based model [62]. In this experiment, the coordinates of bounding boxes are used to generate pixel-wise masks with approximate size and location of defects. Then similarly, a 4-skip U-Net model was trained under the supervision of pixel-wise masks with 70 epochs. Other hyper-parameters such as learning rate, batch size, and weight decay were set to $1e^{-4}$, 16, and $5e^{-5}$, respectively.

The relatively low-performance scores of Setting 3 (top right of Table III) show that even with a considerable amount of defect data, it is still difficult for the model to discover the location information of defects only from label annotations. On the contrary, the results of Setting 4 indicate that even rough spatial information of defect can significantly improve the detection performance for supervised models.

D. Setting 5: Large Amount of Defect Data With Mask Annotations

In Setting 5, there are a large number of defect samples with detailed pixel-wise annotations. As a result, all data-driven algorithms trained under this setting achieved the best detection performance compared with the rest settings. But it should be emphasized that this setting is not the first choice to develop since its prohibitive costs for data collecting and labeling.

Considering the cell images and fabric pictures both featured with homogeneous background and low-level semantic information, we adopted U-Net [34] algorithm, which was initially designed to segment cells on microscopy images, as a defect detection model for this setting. The success of U-Net in cell image segmentation shows the effectiveness of skip connection in keeping fine grained details of deep neural networks [53]. And given the importance of low-level spatial information of defects in real-world practice, U-Net is a suitable baseline model for defect detection in this setting. During this experiment, a U-Net

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with 4 skip connections was trained to perform the detection task. The learning rate, batch size, weight decay, and epochs were set to $5e^{-4}$, $1e^{-3}$, 32, and 200 for the fabric-wise training. And the learning rate, batch size, weight decay, and epochs were set to $1e^{-3}$, $5e^{-5}$, 64, and 150 for the group-wise training and the total-data training.

As shown in the bottom right of Table III, the U-Net fed with mask annotations achieved better detection performance than the rest of methods under other Settings. There is also a comparison between models trained in different ways. The models trained on each fabric and a group of fabrics achieved better performance than the general model (trained on the whole dataset at once). Meanwhile, this result suggests that using more training data may not be helpful for the inspection model. The special nature of fabric inspection task requires the inspection algorithms should be aimed at single or several fabrics with similar patterns, rather than designing a general model to deal with all fabrics, in terms of training efficiency and data consumption. Furthermore, the detection performance continuously improves under every next setting, which in turn validates our approach to distinguish different task settings and establish a uniform evaluation protocol to compare between them.

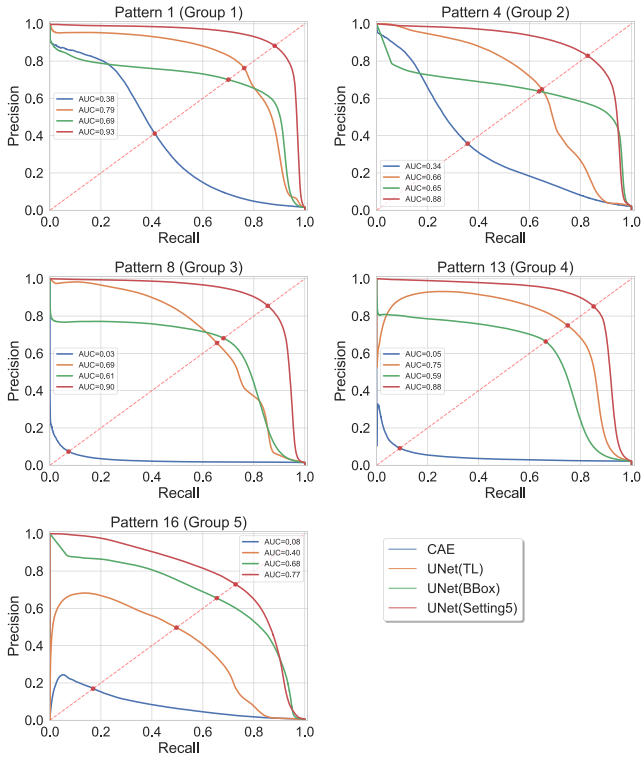


Fig. 7. The pixel-wise precision-recall curves of selected methods on 5 different fabric patterns. The area under the curve (AUC) is commented in the legend and the red dots are break-even points.

E. More Comparisons

In addition to numerical results, we also conducted in-depth comparisons between some methods. In order to save space, we choose only one fabric from each group and one method from each basline experiment to perform the following analysis.

1) Precision-Recall (PR) Curve: Before the model outputs pixel-wise binary mask predictions, we can usually obtain the score maps with continuous real values. These score maps can provide us with more information to analyze methods' behavior. Here, the pixel-wise precision-recall curves (Fig 7) are adopted to demonstrate these methods' capacity for detecting defective pixels. The PR curve shows the tradeoff between precision and recall for different threshold. A high area under the curve (AUC) indicates that the model can achieve both high recall and high precision. The Break-even (BEP) point of PR curve represents a point that precision and recall are equal. The further from the origin the better the performance. As shown in Fig. 7, the results displayed by AUC and BEP are consistent with the proposed metrics, which indicates the rationality of the evaluation protocol.

2) Different Sized Defects: As mentioned in the proposed evaluation protocol, we can further analyze the model's detection performance on different sized defects based on the region-level metrics. Table IV shows a comparison between these models. The results show that these models tend to detect large defects while ignoring small defects, especially for the supervised methods. There are two reasons for this phenomenon. The first is that smaller defects are harder to be perceived on the images, and the detection of small defects is more difficult than large defects. Second, these CNN-based supervision methods use pixel-wise cross-entropy as the loss function for parameter optimization, which causes a similar problem in the pixel-level evaluation, that is, large defects containing lots of pixels will dominate the loss.

TABLE IV
THE REGION-LEVEL EVALUATION RESULTS OF SELECTED METHODS ON THE DEFECTS OF SMALL, MEDIUM, AND LARGE SIZE

Method	Group	$F_1^{reg}{}_s$	$F_1^{reg}{}_m$	$F_1^{reg}{}_L$
CAE	Overall	0.011	0.263	0.284
UNet (TL)	Overall	0.097	0.628	0.776
UNet (BBox)	Overall	0.145	0.816	0.900
UNet (Setting5)	Overall	0.281	0.881	0.931

3) Visualization: Some visual results are useful to help understand the performance of different methods, as Fig. 8 shown. The method CAE without using any defect data (Setting 1) got poor detection performance. At the same time, we can see that even with a small amount of supervision (fewer defects or weak annotations), the defection results can be significantly improved.

V. DISCUSSION AND CONCLUSION

In this work, we mainly present a benchmark fabric dataset that is carefully designed to advance the development of visual fabric inspection by (i) creating a large dataset with many high-quality images, coarse-to-fine annotations, and practical task settings; (ii) developing a sound evaluation protocol with pixel-level, region-level, and sample-level metrics for performance comparison; and (iii) providing baseline experiments to demonstrate some potential solutions and the usefulness of this

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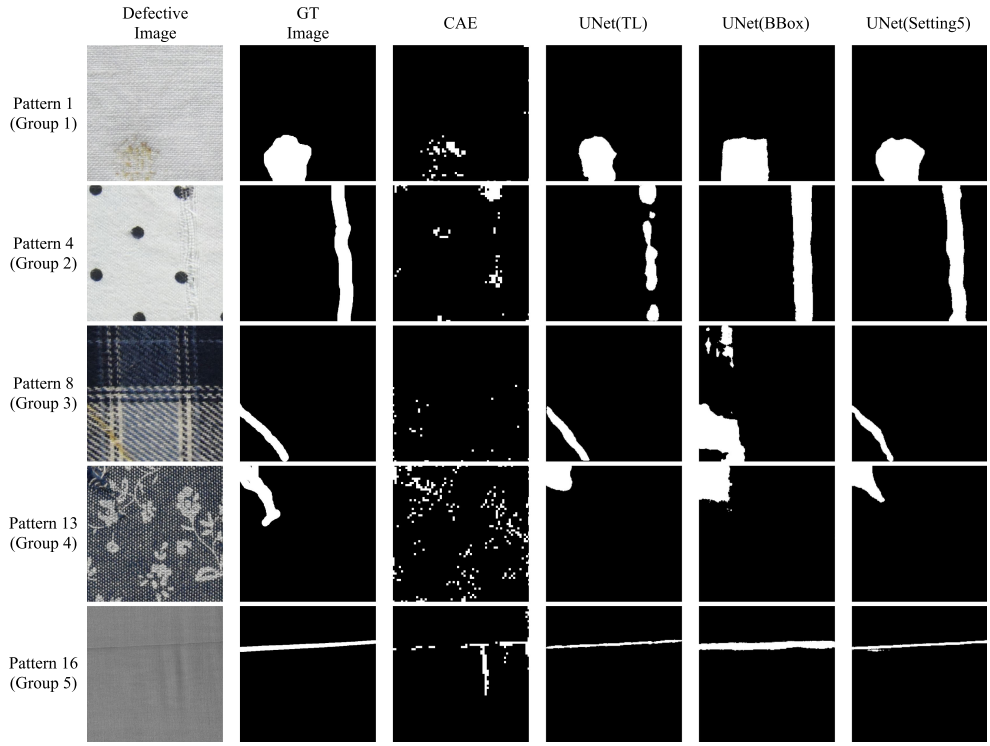


Fig. 8. Some visual results for these baseline methods.

benchmark dataset. And the results, in turn, support the design of task setting structure and evaluation protocol.

In the baseline experiments, we highlighted the potential methods for 5 task settings respectively. As the results show, the unsupervised anomaly detection of image data fails to achieve satisfactory performance in the case of insufficient annotation. Perhaps the saliency detection based on unlabeled samples is a more promising approach. The idea of transfer learning and data expansion can be used to alleviate the shortage of defective samples to some extent. In addition, one-shot or few-shot learning and the method of integrating hand-coded knowledge into the modeling process are also worth exploiting for future research. On the other hand, using coarse to fine annotations to progressively improve the detection performance during the learning process may be a way to bridge the performance gap between unsupervised methods (using only defect-free samples) and supervised methods (using only defective samples). At the same time, these strategies can also be used to speed up the development process of detection algorithms.

Our future work will focus on three goals: 1) Expanding the dataset. We will continue to collect more images with different patterns and defects and make them available to researchers in this field. 2) Collecting a performance benchmark. We will maintain a performance table of the state-of-the-art algorithms tested by the dataset, in order to enable easy comparison between existing and new algorithms. 3) Exploiting new methods. We will research algorithms based on the dataset. Our baseline experiments reveal some possibilities for addressing these challenges, however, the results also suggested that there is still a lot of research space to develop new detection methods.

Although ZJU-Leaper outperforms the peer datasets in many aspects, it is not yet the perfect representation of the real fabric world. Also, all baseline experiments display only a small part of the possibilities for future research. Rather, we expect our work to serve as a foundation from where data can be further enriched and more detection algorithms can be developed, hence promoting the advancement of automated fabric inspection for the textile industry.

APPENDIX
SIZE CONSTRAINTS

Because the size of the defect can be reflected by the number of defective pixels, we estimate the size of each defect through its mask annotation, and dividing all defects into three groups according to two thresholds of 50^2 and 100^2. Fig. 9 shows the proportion of each size group to all defects.

Concretely, the number of proposed regions and the number of ground-truth regions of small, medium and large-sized defects can be calculated based on the following rules:

- small proposed regions $= \{ P \mid \ |P| < 50^2 \}$, small ground-truth regions $= \{ G \mid \ |G| < 50^2 \}$;
- medium proposed regions $= \{ P \mid 50^2 < |P| < 100^2 \}$, medium ground-truth regions $= \{ G \mid 50^2 < |G| < 100^2 \}$;
- large proposed regions $= \{ P \mid \ |P| > 50^2 \}$, large ground-truth regions $= \{ G \mid \ |G| > 50^2 \}$;

where, P/G represent the binary map of proposed region and ground-truth region, and $| * |$ represents the cardinality (i.e. the number of defective pixels) of a region.

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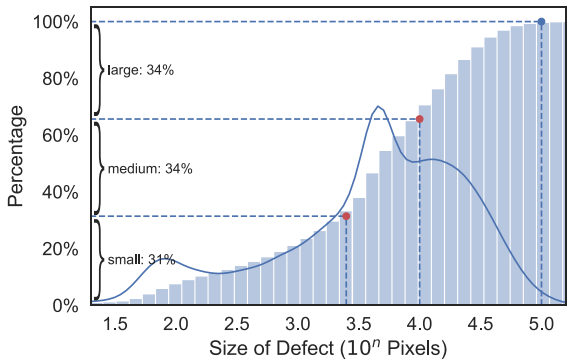


Fig. 9. The distribution and cumulative histogram of defect size of ZJU-Leaper. According to the size thresholds of 50^2 and 100^2, all defects can be divided into small, medium, large groups, which account for 31%, 34%, and 34% of all defects respectively.

TABLE V
THE EVALUATION RESULTS OF 5 IMPLEMENTED ALGORITHMS DESIGNED FOR THE HKU-FABRIC DATASET. ALL EXPERIMENTS ARE CARRIED OUT ONLY ON THE FABRICS OF GROUP 1

Method	Group	F_1^{pix}	F_1^{reg}	F_1^{sam}
lattice Segmentation [43]	Group1	0.035	0.038	0.396
Low-rank Decomposition [44]	Group1	0.380	0.421	0.998
Elo Rating [41]	Group1	0.154	0.106	0.698
Image Decomposition [40]	Group1	0.034	0.015	0.400
Regular Band [14]	Group1	0.038	0.009	0.402

OTHER IMPLEMENTATIONS

We also implemented 5 other inspection methods on Setting 1, some of which [14], [40], [41] are typical methods which are widely cited as baselines in many articles, and others [43], [44] claimed to be the state of the art. However, we found that these methods are highly engineered for HKU-Fabric dataset, and it is difficult to adapt these models to ZJU-Leaper with more complex textures. Therefore, these methods were evaluated only on Group 1 which only consists of simple textures. More details about these implementations could be found in the public code repository.

For Table V, there are some explanations about these implemented methods. First, the latest lattice segmentation method performed worst since it is reported only workable for fabrics with horizontally and vertically repeated patterns [43]. Second, inspection algorithms of image decomposition and Elo rating obtained low scores because they need to adjust many hyper-parameters for each fabric separately, and we only used the same set for all fabrics during the experiment. Only the low-rank decomposition method achieved good results, but it should be noted that it was only applied to defective samples, so the performance of normal images is excluded. Third, the results we obtained are not as good as they reported, partly because the performance of these methods depends on a few hyperparameters which have not been given explicitly in the papers. In summary, the unsatisfactory results of these methods on more complex patterns highlight the room for improvement in defect inspection methodology under this low-cost setting (Setting 1).

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